traverse both branches, and check spheres S_3 to S_6 for intersection with Z.

To continue our images dataset example, suppose that now we would like to check if a specific image Z is in our dataset: one option would be comparing Z to all images in the dataset. Comparing two images might require some time (especially if, for instance, all images has the same size, and we can't do a quick check on any other trivial image property to rule out obviously different pairs); recalling that our dataset supposedly has ten of thousands of images, if we go this way, we should be prepared to take a long coffee break (or, depending on our hardware, leave our machine working for the night).

Listing 10.3 The search method

```
function search(node,
target)                                                    #1
  if node.leaf
then
#2
    for each point in node.points
do                                                  #3
      if point == target then
        return
node
#4
  else
    for each childNode in node.children
do                                          #5
      if childNode.intersectsPoint(target)
then                                      #6
        result ← search(childNode,
target)                                         #7
```